

Aislyn Bell

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Education

Johns Hopkins University
Ph.D. Physics and Astronomy

(August 2025-May 2031)

University of Colorado at Boulder
BA Astronomy - Astrophysics *cum laude*

(August 2020 - May 2024)

Research Experience

Johns Hopkins University
Graduate Research Assistant

(August 2025-Present)

- Advisor: Meredith MacGregor
 - Running N-body simulations to interpret the dynamical history of debris disks and the undetected planets therein
 - Co-I of Nancy Grace Roman Space Telescope proposal to search for transiting exocomets in the Galactic Bulge Time Domain Survey and High-Latitude Time Domain Survey

Space Telescope Science Institute
Graduate Research Assistant

(August 2025-July 2026)

- Advisor: Leonardo dos Santos
 - Modeled the photoevaporation rate on exoplanets from archival JWST/SOSS observations using hydrodynamic escape codes like ATES/p-winds

University of New Hampshire
Undergraduate Research Assistant

(May 2022-July 2025)

- Advisor: Lynn Kistler
 - Measured and compared ion outflow for different methods of identifying substorm onsets from TEAMS data on the FAST Spacecraft.

Harvard-Smithsonian Astrophysical Observatory
Undergraduate Research Assistant

(June 2023-May 2024)

- Advisors: Jennifer Yee and Weicheng Zang
 - Co-I on Las Cumbres Observatory Global Telescope Network Key Proposal. Observed and modeled high-magnification microlensing events to yield planet detections.

Refereed Publications

- “KMT-2023-BLG-1431Lb: A New $q < 10^{-4}$ Microlensing Planet from a Subtle Signature,” Aislyn Bell *et al* 2024, *PASP* **136** 054402
- “Coronal Heating as Determined by the Solar Flare Frequency Distribution Obtained by Aggregating Case Studies,” James Paul Mason *et al* 2023 *ApJ* **948** 71

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Contributed Talks

- “Constraining Hidden Planets in the Fomalhaut Debris Disk Using Both Belts”, ERES XI, University of Michigan, July 2026
- “New Constraints on Hidden Planets in Fomalhaut from Both Belts”, Debris Disk Connections, University of Cambridge, July 2026
- “Variation of Auroral Ion Outflow During Substorms,” AGU December 2022

Observing

- **Las Cumbres Observatory Global Telescope Network** (56 hours total): photometry of microlensing exoplanets
 - 48 hours in 2023A, 8 hours in 2023B
- **Sommers-Bausch Observatory** (18 nights total): Outreach, occultations, and spectroscopy of stars
 - **20”**: 8 nights
 - **24”**: 10 nights
- **Apache Point Observatory 3.5 m DIS Spectrograph** ($\frac{1}{2}$ night total): Confirming the age of young stars

Awards and Fellowships

- **Observatory Fellow**, Maryland Space Grant Observatory Consortium (2027)
- **Graduate Research Fellowship Honorable Mention**, National Science Foundation (2026)
- **Esteemed Scholars Scholarship for academic excellence**, CU Boulder (2020-2024)
- **Susan E. Wesley Undergraduate Scholarship**, CU Boulder (2023-2024)
- **J. Tour Scholarship in Arts & Sciences for outstanding rising seniors**, CU Boulder (2023-2024)

Selected Work and Outreach Experience

Co-President of Organization for Gender Minorities and Women in Physics (July 2026-Current)

Johns Hopkins University

- Organizing the Annual Summit, a 1 day conference highlighting research by gender minorities in physics, facilitating networking, and building community
- Planning annual spring brunch in collaboration with Morgan State University to facilitate cross-institutional relationships
- Coordinate with GWiP board to run social programs for undergraduate students, graduate students, and postdocs
- Manage organization budget

Aerospace Educator

(August 2024-May 2025)

Wings over the Rockies Air and Space Museum

- Taught 5 middle school classes and 2 high school classes on astronomy
- Created curriculum for my 7 classes, and for programming classes taught by other instructors
- Created an outreach program for the inflatable planetarium, debuting with 2 shows of my creation
- Assisted at various outreach events to promote and grow the education program at the museum
- Interviewed notable astronomer Jocelyn Bell Burnell on the Behind the Wings podcast

Student Assistant and Outreach Coordinator

(March 2023-May 2024)

Sommers-Bausch Observatory

- Assembled, calibrated, and tested labs for introductory astronomy courses

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- Operated the heliostat, 20", and 24" telescopes
- Assisted in building, maintaining, and upgrading hardware for the 24" and 12" telescopes, and the heliostat
- Assisted in calibrating the 20" telescopes
- Scheduled field trips, private observing events, and outreach events
- Assisted in organizing and running annual outreach events that 400+ people attend
- Created new outreach events to increase DEI
- Substituted for undergraduate learning assistants as needed

Resident Advisor

(August 2021-May 2022)

CU Boulder

- Participated in an on-call rotation to complete safety rounds of the building and address resident concerns
- Implemented curriculum on diversity, inclusion, and identity development for 400 first-year college students
- Mediated interpersonal conflicts and supported residents in crisis by providing immediate referrals and interventions
- Facilitated a welcoming and inclusive environment for students, staff, and visitors, promoting positive interactions and community development within the residence halls